

QUERYQUEST ASSIGNMENT – 01

PART 01:

Q1. A traditional file system stores data in separate files that are managed independently. Whereas, a Database Management System (DBMS) stores data in a centralized database. A DBMS reduces data duplication.

A university student records system is a good example where a DBMS is better. Since multiple departments need access to the same student information, a DBMS ensures that everyone sees the same updated data and helps avoid storing duplicate records.

Q2. A strong entity can be uniquely identified using its own primary key, whereas a weak entity cannot. The primary key of a weak entity is formed by combining its partial key with the primary key of its owner entity. The relationship that connects a weak entity to its owner entity is called an identifying relationship.

Q3. Phone number should be modeled as a multivalued attribute because a student can have more than one phone number. A simple attribute can store only one value, while a multivalued attribute allows multiple values for the same entity.

Q4. A super key is any set of attributes that can uniquely identify a row in a relation. A candidate key is a minimal super key, meaning no attribute can be removed from it.

Candidate Keys: {roll no}, {email}

Super Keys that are not candidate keys: {roll no, name}, {email, branch}

Q5. The referential integrity constraint (foreign key constraint) is violated. It prevents records from referencing data that does not exist in another table. The insert operation should be rejected because the student ID does not exist in the Student table.

Q6. A junction table must be created. Its primary key consists of the primary keys of the participating entities combined together. It cannot be represented as a single column because a record may be related to multiple records in the other table. A junction table is needed to store these multiple relationships.

PART 02:

PART A – ENTITY AND RELATIONSHIP IDENTIFICATION

Entities and Key Attributes:

Customer

- CustomerID (Primary Key)
- Name
- PhoneNumber

Restaurant

- RestaurantID (Primary Key)
- RestaurantName
- Address

Order

- OrderID (Primary Key)
- OrderDate
- TotalAmount

Menu Item

- ItemID (Primary Key)
- ItemName
- Price

Relationships

1. Customer places Order

Cardinality: 1:N

A customer can place multiple orders, but each order is placed by only one customer.

2. Restaurant receives Order

Cardinality: 1:N

A restaurant can receive many orders, but each order is associated with only one restaurant.

3. Restaurant offers Menu Items

Cardinality: 1:N

A restaurant can offer multiple menu items but each menu item belongs to only one restaurant.

4. Order contains Menu Item

Cardinality: M:N

An order can contain multiple menu items and the same menu item can be in many different orders.

PART B – ATTRIBUTE CLASSIFICATION

One Multivalued Attribute – Phone Number (Customer)

Phone Number is a multivalued attribute because a customer may have more than one contact number.

One Composite Attribute - Address (Restaurant)

Address is a composite attribute because it can be divided into smaller components such as street, city, state, and PIN code.

PART C – RELATIONAL SCHEMA DESIGN

Relational Schema:

OrderMenuItem(OrderID, ItemID, Quantity)

Primary Key: (OrderID, ItemID)

Foreign Keys:

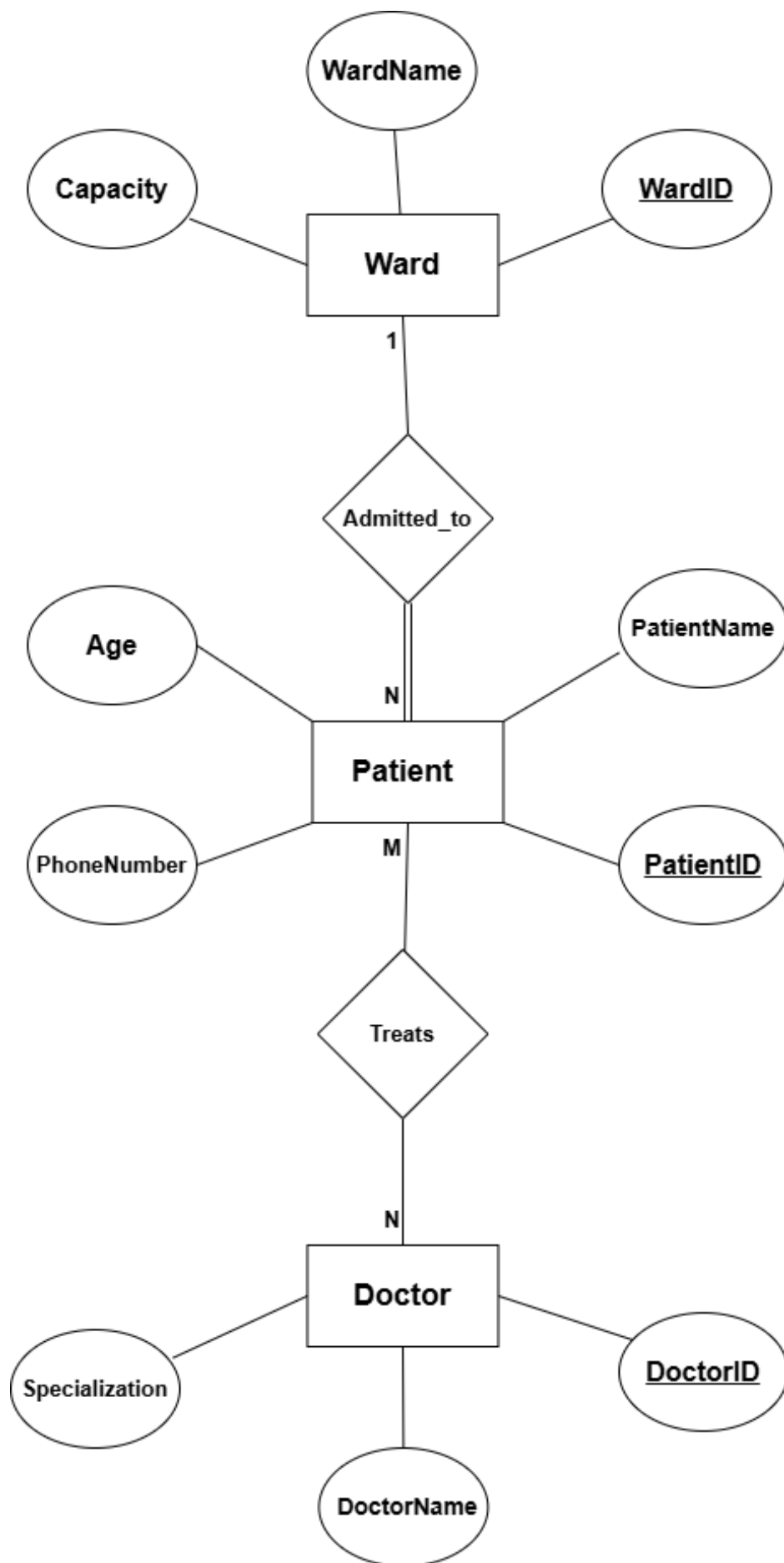
- OrderID references Order(OrderID)
- ItemID references MenuItem(ItemID)

Additional Attribute:

- Quantity – stores the number of units of a menu item included in an order

PART 03:

ER DIAGRAM DESIGN :



DESIGN JUSTIFICATION :

The relationship between Doctor and Patient is many-to-many (M) because a doctor can treat multiple patients and a patient can also be treated by multiple doctors.

In the Admitted_to relationship, Patient has total participation because every patient in the system must be assigned to a ward. This is shown using a double line between Patient and Admitted_to in the ER diagram. Ward has partial participation because a ward can exist even when no patients are currently admitted to it. In the Treats relationship, both Doctor and Patient have partial participation. A doctor may be present in the system before treating any patients, and a patient may be admitted to the hospital before being assigned a doctor.

For the attributes, I included PatientName, DoctorName, and WardName as simple attributes because they each store a single piece of information. I also used PatientID, DoctorID, and WardID as primary keys to uniquely identify each entity in the database.